Unit 1 – Using Objects and Methods Name: **KEY**
HW 5

1) Identify the following parts of a completed class below.

```
public class Movie {

    private String title;
    private int runTime;

    public Movie() {
        title = "Unknown";
        runTime = 1;
    }

    public Movie(String t, runTime r) {
        title = t;
        runTime = r;
    }

    public void displayMovie() {
        System.out.println("Title: " + title);
        System.out.pritln("Runtime: " + runTime);
    }

}
```

a) Instance Variables

b) Default Constructor

c) Parameter Constructor

d) Method

2) Using the class above,

a) Write a statement that creates a movie object using the default constructor:

```
Movie barbie = new Movie();
```

b) Write a statement that creates a movie object using the parameter constructor:

```
Movie harryPotter = new Movie("Prisoner of Azkaban", 142);
```

_____ 3) Which of the following best describes the relationship between a class and an object?

(A) A class is a specific variable in a program, which has behaviors defined by objects.
(B) A class is a piece of data which is contained in an object.
(C) An object is used to create classes, which are primitive data types.
(D) A class is a template or blueprint from which an object can be created.

_____ 4) Which of these code segments shows a constructor being used to create a new object?

(A) `System.out.print("new Dog");`
(B) `Dog doggie;`
(C) `double num = 2.65;`
(D) `Dog doggie = new Dog();`

_____ 5) Which of the following is the general schematic for a class definition?

(A)
```
public class ClassName {
     //Instance Variables
     //Constructors
     //Methods
}
```

(B)
```
private class ClassName {
     //Instance Variables
     //Constructors
     //Methods
}
```

(C)
```
Class ClassName {
     //Instance Variables
}
```

(D)
```
public class ClassName {
     public static void main(){
          //Instance Variables
          //Constructors
          //Methods
     }
}
```

7) Write "T" for true and "F" for false on the line next to the statements below.

a) __T__ A class is a template for an object

b) __T__ Encapsulation is protecting data from being manipulated by methods and classes that you don't want to change the data

c) __F__ `Dog doggie;` is an example of instantiation

d) __T__ You can only have one default constructor in a class

e) __F__ You can only have one parameter constructor in a class

f) __F__ The keyword `private` means that the method or data can be accessed by any class that uses a reference to the object it is in.